SOLPS-ITER download and installation instructions

Welcome to SOLPS-ITER!

The first step, after you have obtained an ITER IDM account, is to agree to the IMAS terms of use and request access to the needed repositories. The form to fill out for this is to be found at: https://jira.iter.org/servicedesk/customer/portal/1/create/257. This may also require the signature of a Research Collaboration Agreement between the IO and the requesting user's institution, if one is not already in force. For a full SOLPS-ITER installation, you should request access to the following repositories:

- SOLPS-ITER
- B2.5
- EIRENE
- CARRE
- DIVGEO
- TIARA
- SOLPS-GUI
- GR Software
- MSCL
- AMNS ADAS

You may request access to other repositories as well according to your needs or interest. If you do not intend to participate in code development, you can ask for REPO_READ permissions only, instead of REPO_WRITE.

You can generally browse the ITER Git repositories here: https://git.iter.org and SOLPS-ITER specifically here: https://git.iter.org/projects/BND/repos/solps-iter/browse, but since these repositories are restricted you will first need to authenticate with your ITER username and password – possibly twice, once for the intranet, and once for the Git web front end in the top right-hand corner.

If you upload a copy of your public *ssh* key at https://git.iter.org/plugins/servlet/ssh/account/keys or create a token at https://git.iter.org/plugins/servlet/access-tokens/manage, (by following links under “Manage Account”), then you should be able to simply clone a copy of SOLPS-ITER to your local machine by doing:

```
> git clone ssh://git@git.iter.org/bnd/solps-iter.git
> cd solps-iter
> git submodule init
> git submodule update
```

If you want to clone SOLPS-ITER into a different directory than the “solps-iter” default name, replace the initial “git clone” command by:

```
> mkdir <my_personal_SOLPS-ITER-directory_name>
> cd <my_personal_SOLPS-ITER-directory_name>
> git clone ssh://git@git.iter.org/bnd/solps-iter.git .
```

Please note that because of the way some of the SOLPS-ITER utility scripts function, the path to the SOLPS-ITER top directory must not contain any of the strings “.debug”, “.mpi”, “.openmp”, “.ig”, “.adj” or “.tgt”.

Since your user ID is immutably encoded into any changes, it's always a good idea to properly configure your local installation of Git on all the systems you plan on making code revisions on with, e.g.:

> git config --global user.name "Your Name"
> git config --global user.email your.name@where.you.are

It is currently recommended that you position yourself on the *master* branches in each of the submodules, by means of the *switch_to_master* script (which can be used after you establish your SOLPS-ITER session environment, see below), or by doing the following:

> git checkout master
> cd modules/B2.5
> git checkout master
> cd ../Eirene
> git checkout master
> cd ../DivGeo
> git checkout master
> cd ../Carre
> git checkout master
> cd ../adas
> git checkout master

If you are planning to do code development, your starting point should be the *develop* branch, which you can upload by using "*solps-iter_update_develop*", after first checking out that branch at the SOLPS-ITER top level and rehashing.

When updating the code in this way, the script will check if you have any local branches where you have done some local work (or if your local work is not yet on a branch), and, if you have not done so already, ask you to commit or stash it first before proceeding. After the update is completed, you will be able to rebase your local work on top of the updated code as necessary. Please check the *README* file in your *$SOLPSTOP* directory for further instructions on how to safeguard your work and update your code version.

The *solps-iter_update* script automatically recompiles the standard version of the code (no debug, no MPI, no OpenMP) but also updates the dependencies lists for all versions. If you wish to recompile others versions of the updated code, you can then do:

> gmake {your favourite target[s] here}

It is possible to pass Makefile instructions, in particular multi-threading, with the syntax:

> gmake MAKE_OPTIONS=-j8 solps

It is not recommended to pass such instructions directly to the Makefile (say using "*gmake –j8 solps*") because the SOLPS-ITER compilation is rather complex and must be done in a certain order with some parts required to be serial.

Users who have obtained REPO_WRITE permissions can, for their own development, if any, create *$USERNAME/{feature/[topic]|bugfix/[topic]|hotfix/[topic]}* branches and raise a Pull Request (https://git.iter.org/projects/BND/repos/solps-iter/pull-requests) to have them merged into the develop branch after pushing them back to the ITER repository, where *$USERNAME* is their ITER IDM account name. This naming convention is meant to allow the SOLPS-ITER ROs to more easily manage everybody's branches. Please conform to it. The *develop*, *master*, and *svn/** branches are protected and can only be manipulated by the owners of the repository.

Issues are handled using JIRA (https://jira.iter.org/browse/IMAS) and you should feel free to create new issues associated with SOLPS-ITER or any of its submodules (selected from the drop-down list of components) if you have any problems or suggestions. (You can ignore the options associated with resource estimates and sprints.)

For help with Git, we can recommend the documentation here (http://git-scm.com/) and the Pro Git book (linked from there) in particular. Another useful resource for novice Git users is the tutorial from Codecademy available at https://www.codecademy.com/learn/learn-git.

The documentation specific to SOLPS-ITER can be found at:
https://sharepoint.iter.org/departments/POP/CM/IMAS/SOLPS-ITER
and it is visible from here:
https://imas.iter.org/ (then the "*Physics Components"* tab on the left, and choose *"SOLPS-ITER"* from the drop-down list, then follow the link to the official SOLPS-ITER documentation) and where you will also find a manual on how to raise JIRA issues and pull requests and the like.

In addition, there is an IDM folder, located at https://user.iter.org/?uid=Q92BAQ , which contains links to many SOLPS-ITER related documents (those subject to document control procedures and/or presentations about aspects of the code to entities external to the IO, as well as related contract reports).

There is also a Confluence page at https://confluence.iter.org/display/IMP/SOLPS-ITER where contributions from users are welcome, as well as a Slack discussion group which you can join by invitation. Send a request to xavier.bonnin@iter.org to be invited.

Lastly, the full documentation for use of the SOLPS-GUI can be consulted here:
https://static.iter.org/imas/assets/solps-iter/html/ .

Once you have cloned the SOLPS-ITER Repository, you may need to modify the configuration files needed so the necessary environment variables are properly defined. Please check the *README* file and the SOLPS-ITER User Manual (you can use the documentation link above) for details. These include your domain hostname, defined with the "whereami" script, your preferred compiler, in the "default_compiler" script, the modules you wish to load and your preferred debugger name (given with the *DBX=debugger_name* command) in *SETUP/setup.csh.$HOST_NAME.$COMPILER*, and the location and linking instructions for the various libraries in *SETUP/config.$HOST_NAME.$COMPILER*. Common instructions for a given compiler that are not host-dependent can be found in *SETUP/config.common.$COMPILER*. Feel free to look at the examples already present for inspiration. For hosts and compilers on which SOLPS-ITER has already been installed successfully and we know about, we have tried to include these files already in the distribution, but they may be mistaken or incomplete, so please check. In cases where the *HOST_NAME* is not correctly deduced from the “whereami” script, you can specify your desired *HOST_NAME* value by means of a *SETUP/setup.csh.HOST_NAME.local* file, which should contain the line:

setenv HOST_NAME <your_desired_HOST_NAME>

Once these variables are set, to start a SOLPS-ITER session (every time), from the SOLPS-ITER top directory, you do:

```
> tcsh
> source setup.csh
```

Then, if this is the first installation in your file space, please do:

```
> first_setup
```

This will, if they are not already present in the distribution, create the various configuration files for each of the SOLPS-ITER submodules. Feel free to modify them afterwards according to your local requirements. We strongly recommend that you do any modifications you may need in local copies of these files, to be named, respectively: *SETUP/setup.csh.$HOST_NAME.$COMPILER.local* and *SETUP/config.$HOST_NAME.$COMPILER.local*. Once you have made the proper modifications, if these are worth sharing with other users on similar platforms, please copy them other into the original files and commit them back to the repository using a pull request.

Then, if all was filled in properly, a simple “gmake” compilation command (and some patience) will get you a ready-to-run code. You should run the compilation from the SOLPS-ITER top directory, because this is where all the compiler flags are defined and passed down to each of the submodules. The top directory *Makefile* will also build the SOLPS-ITER Manual, which will be placed in *SOLPS-ITER/doc/solps/solps.[pdf|dvi|ps]*.

For the libraries required at compilation (see the full list in Section 1.2 of the SOLPS-ITER Manual), we recommend these be loaded as modules, which can then be shared across multiple users on a single machine, to make the SOLPS-ITER distribution lighter. With the exception of the GR, MSCL and libsonnet libraries, the libraries are not part of the SOLPS-ITER repository, as the IO does not have distribution rights for them. Libsonnet is built automatically from the SOLPS-ITER *Makefile* (although you may need to provide a machine architecture environment variable value). GR and MSCL can be obtained from the ITER GIT server at:

https://git.iter.org/projects/LIB/repos/gr-software/browse
https://git.iter.org/projects/LIB/repos/mscl/browse

When compiling the GR and MSCL libraries, you will need to set two environment variables, in the following way (provided you have already loaded the SOLPS-ITER environment as above):

```
> setenv OBJECTCODE $COMPILER
> setenv SOLPS_LIB $SOLPSLIB
```

For the others, most are freely available from the Internet, with the exception of the NAG library, for which a pay licence is required. However, the NAG library is not strictly necessary to build SOLPS-ITER. And, if you already have an older version of SOLPS (5.x or 4.x), as distributed by IPP-Garching, you may copy the libraries from it into the *SOLPS-ITER/lib/$HOST_NAME.$COMPILER* directory. Local installation of these libraries is the responsibility of the users, and they should seek assistance from their local IT support staff, not from the IO which has neither the resources nor the capabilities to help in that respect beyond what is already provided in the distributed configuration files.

For easier building of the `gfortran` and `ifort64` toolchains, including the IMAS modules and tools for SOLPS-ITER, a *SETUP/easybuild-local.sh* script is provided. The script includes a complete list of modules as they appear on the ITER SDCC cluster and is convenient for standalone/local "builders" and for site administrators that want to build an ITER-equivalent installation on their machine. Example *setup.csh.UL.gfortran* and *setup.csh.UL.ifort64* are thus very similar to *setup.csh.ITER.gfortran* and *setup.csh.ITER.ifort64*, respectively. The configuration files *config.UL.gfortran* and *config.UL.ifort64* can then simply be symlinked to the corresponding *config.ITER* versions.

Essentially, this script exploits the functionality of the [EasyBuild]() tool by adding search paths to ITER-specific easyconfig `.eb` sources from the ITER Git repository. The only requirement for this script is a recent version of Python 3 and `modulesfiles` or `Lmod` support. The rest is being downloaded from the internet and the ITER Git website. A quick start for building may be by creating a Personal Access Token from the ITER Git website as indicated above and entering:

```
> export EASYBUILD_MODULES_TOOL=Lmod # EnvironmentModules are default
> export HTTP_AUTH_BEARER="ReplaceWithPersonalAccessToken"
> SETUP/easybuild-local.sh --help | more
> SETUP/easybuild-local.sh
> SETUP/easybuild-local.sh --imas-foss install
> SETUP/easybuild-local.sh --imas-apps
```

Note that the above minimal example requires Python 3.8+, `Lmod` and Korn shell to be installed on the system and functional. The rest is installed by the script in the *easybuild.local* directory or elsewhere if desired.

If you have any questions about these instructions or encounter difficulties, feel free to contact the SOLPS-ITER RO at:
xavier.bonnin@iter.org